
HOW THE SYSTEM WORKS. IN PICTURES.

Twelve diagrams covering the Phase 1 scope — from the big picture for the venue to the data model for the build team. Companion to the Scope of Work v1.0.

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FOR THE BUILD TEAM

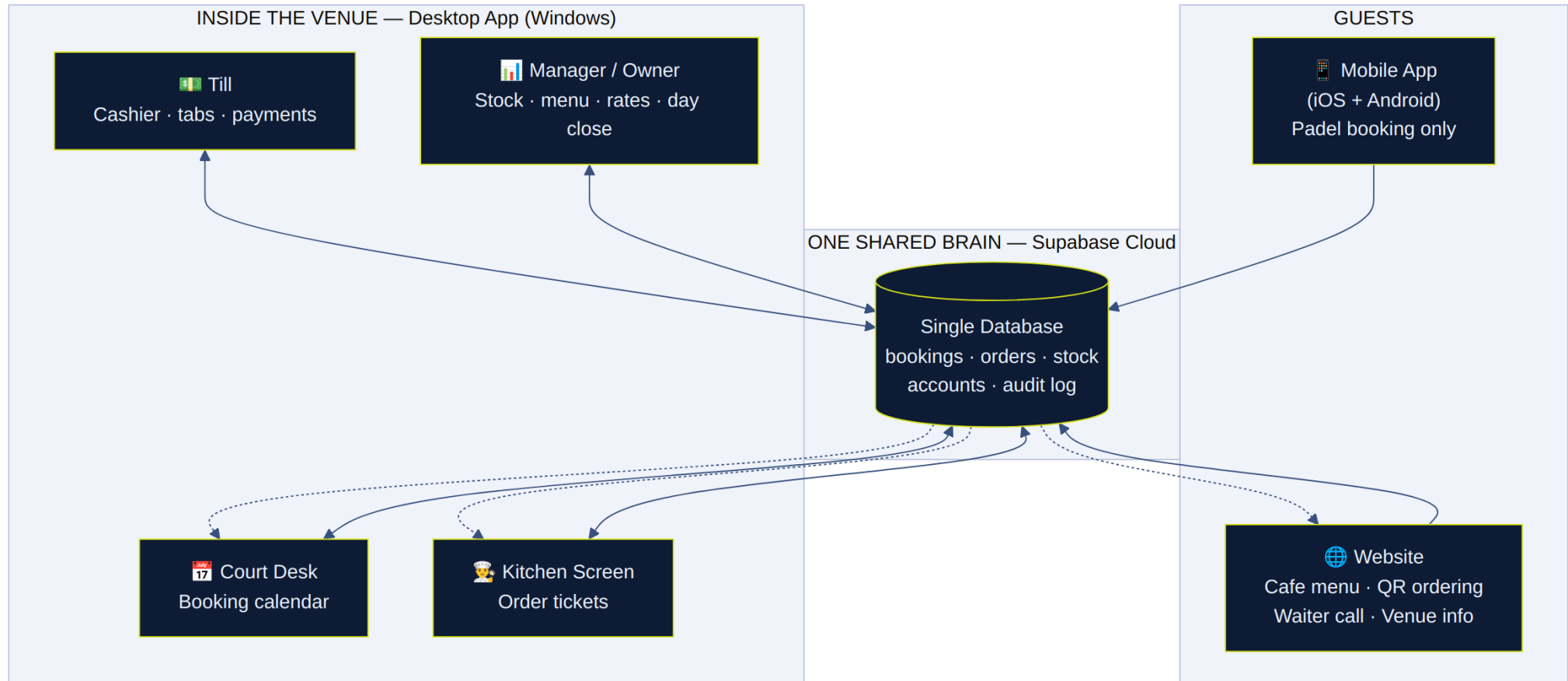
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1 · The System at a Glance

AUDIENCE: EVERYONE

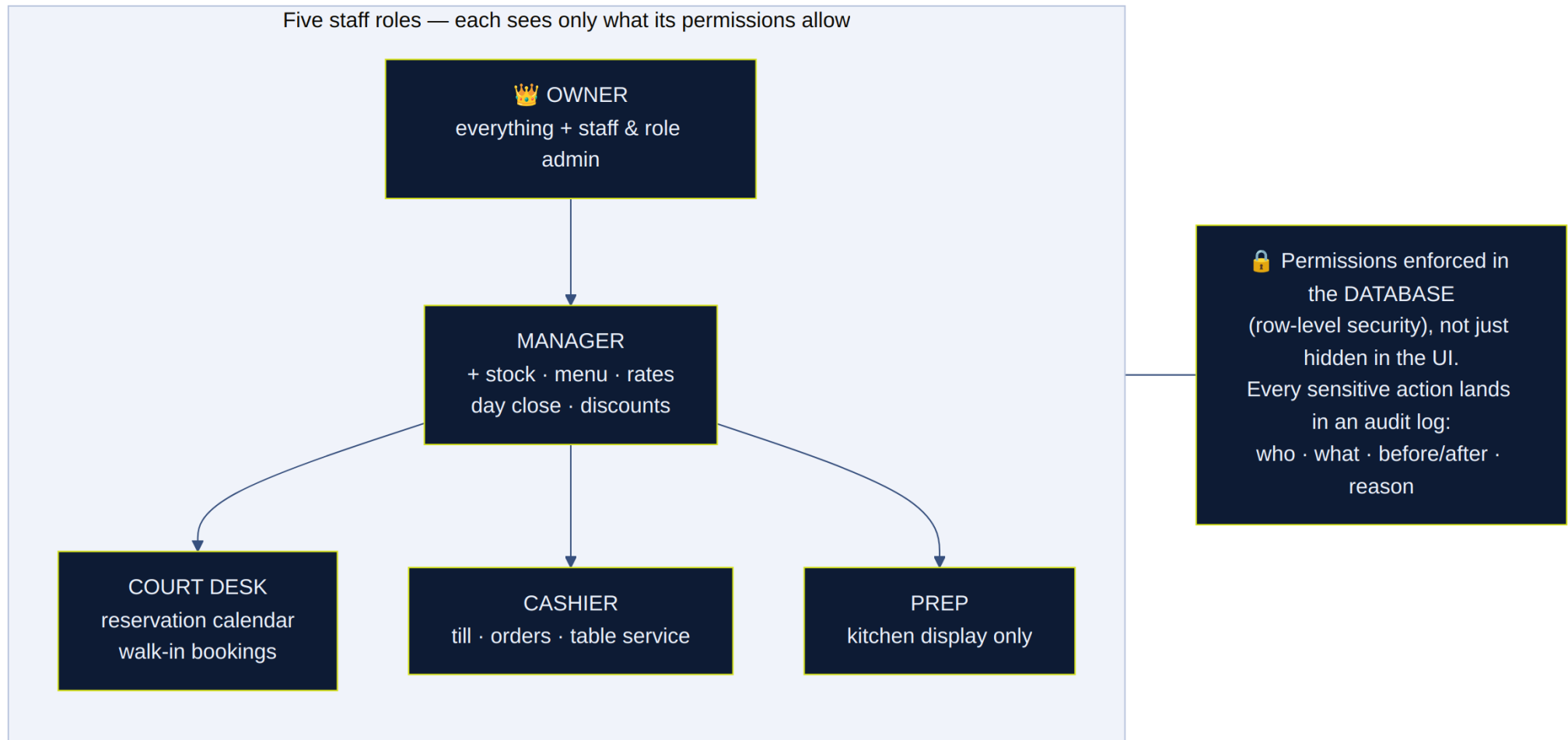


WHAT THIS SHOWS

Three apps, one shared database. Guests use the mobile app for courts and the website for the cafe. Staff use one Windows desktop app whose screens depend on the person's role. Everything reads and writes the same records, and changes push out live — no refreshing.

2 · Who Sees What — The Five Roles

AUDIENCE: EVERYONE

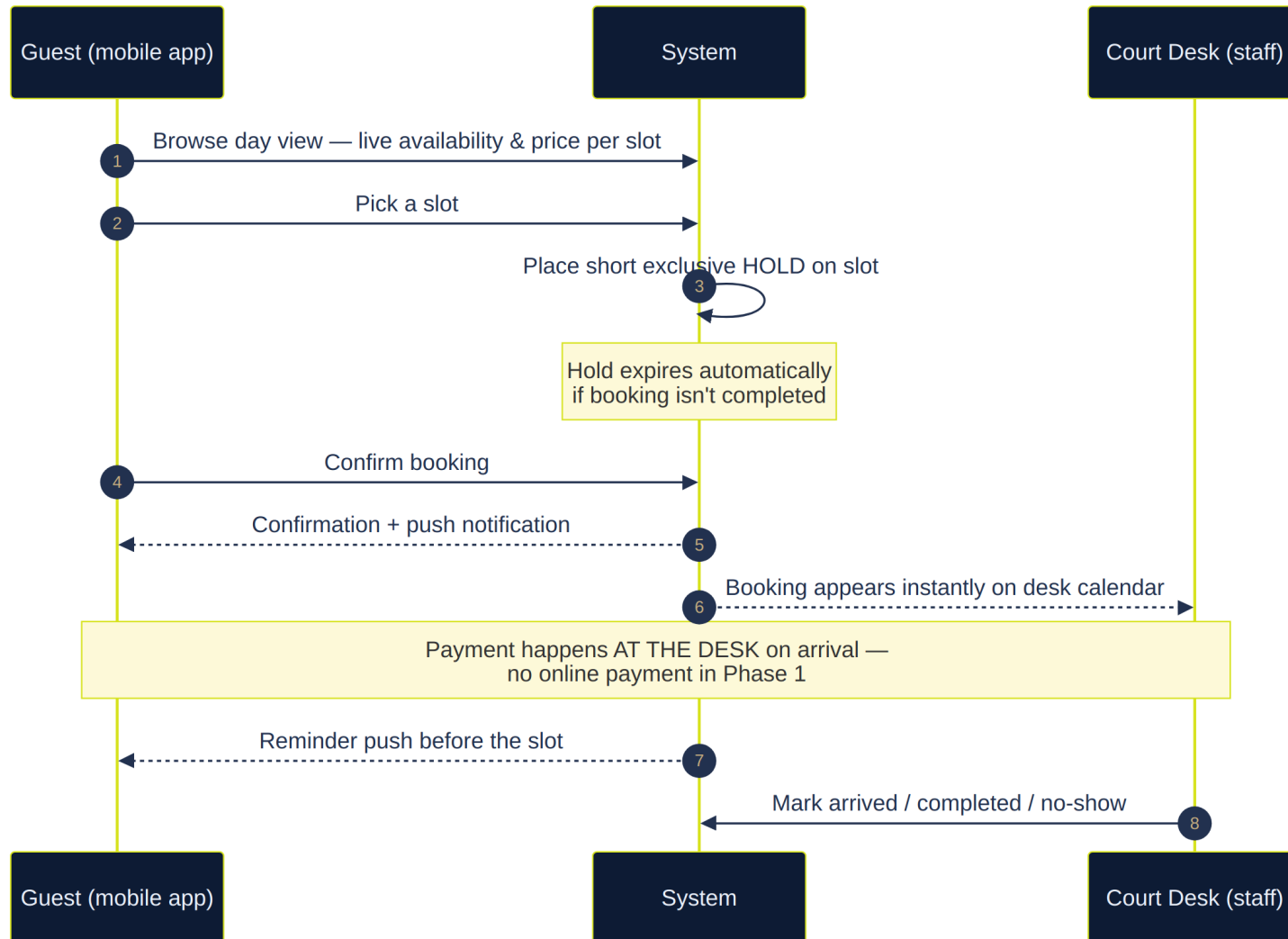


WHAT THIS SHOWS

Each staff role sees only what its permissions allow, and the rules live in the database itself — they can't be bypassed by opening a different screen. Discounts, voids, overrides and stock adjustments all land in a permanent audit log with the person, the change, and the reason.

3 · How a Guest Books a Court

AUDIENCE: EVERYONE

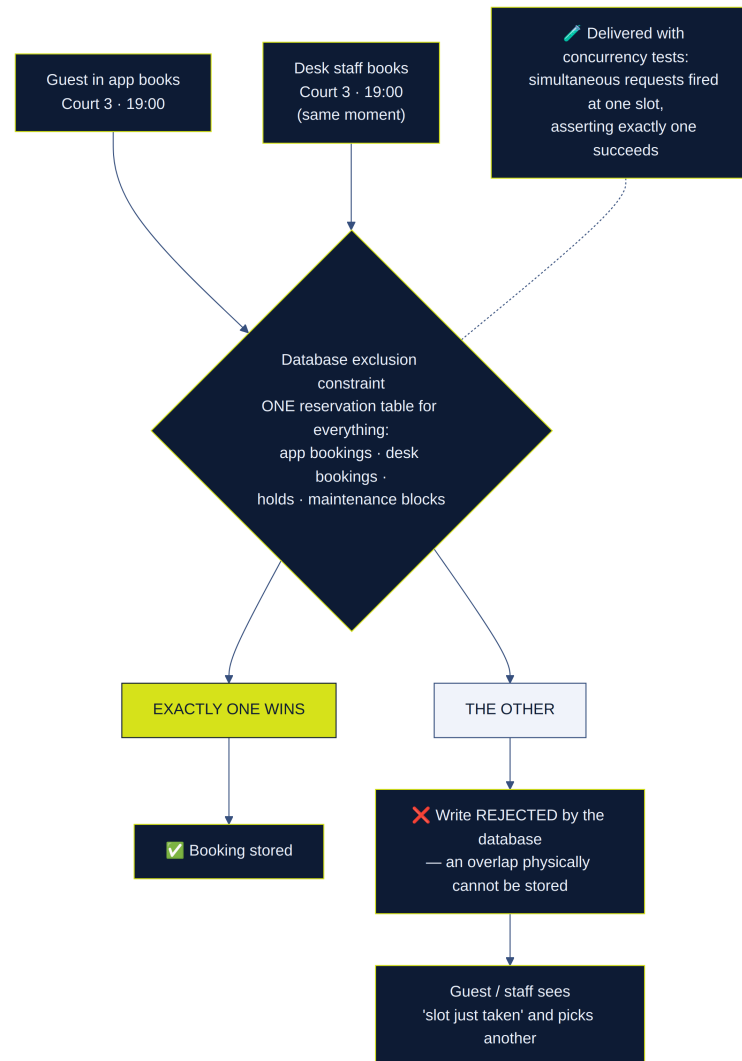


WHAT THIS SHOWS

A slot is briefly held while the guest completes the booking, so nobody else can grab it mid-checkout. The moment it's confirmed, it appears on the desk calendar. Payment stays at the desk — there is no online payment in Phase 1.

4 · Why Double-Booking Is Impossible

AUDIENCE: EVERYONE · KEY GUARANTEE

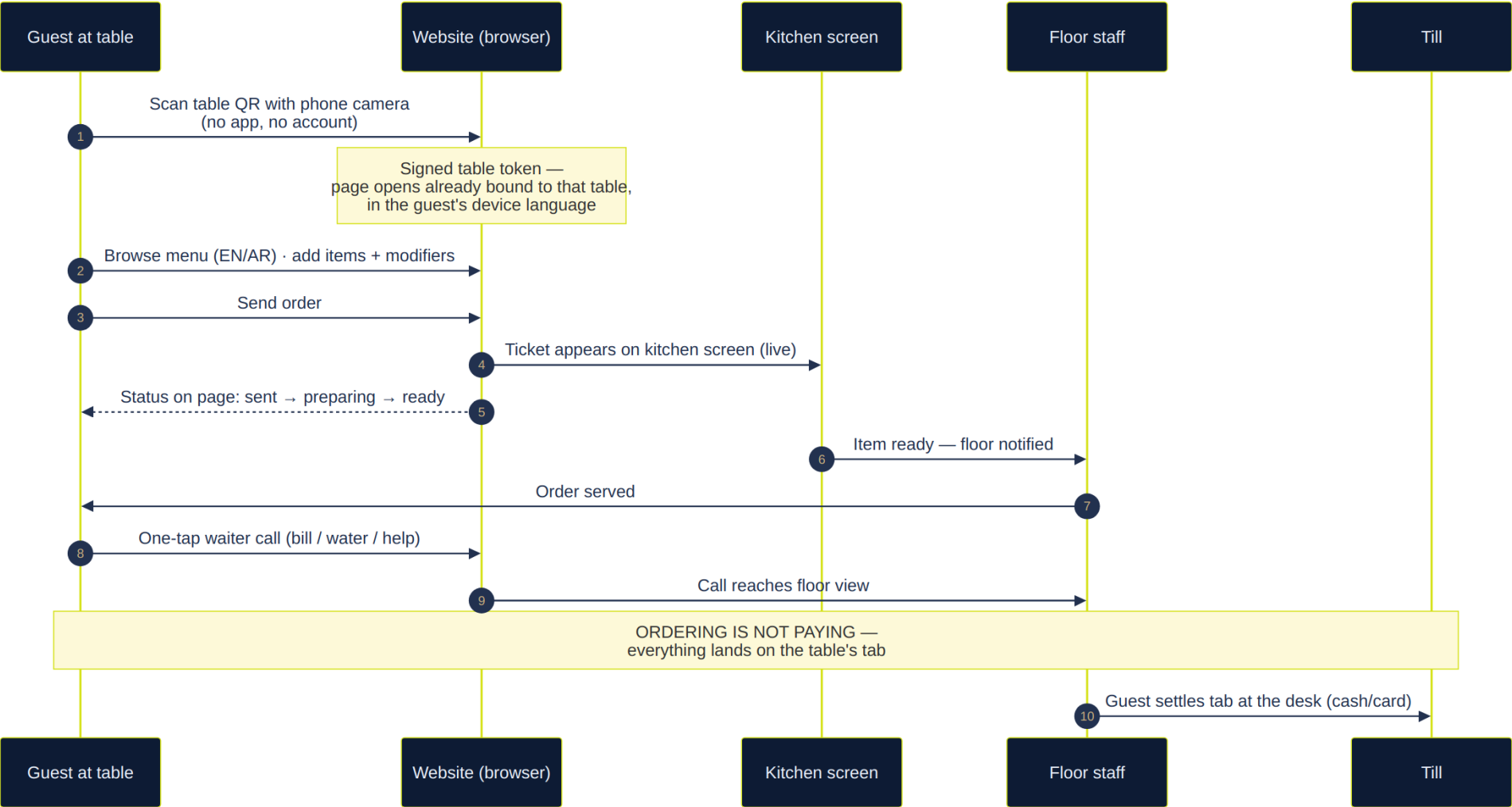


WHAT THIS SHOWS

All court time — app bookings, desk bookings, holds, maintenance — goes into one table the database itself refuses to overlap. A clash isn't a bug we hope to catch; it's a write that physically cannot be stored. Proven by automated tests firing simultaneous requests at the same slot.

5 · How a Guest Orders at the Cafe

AUDIENCE: EVERYONE

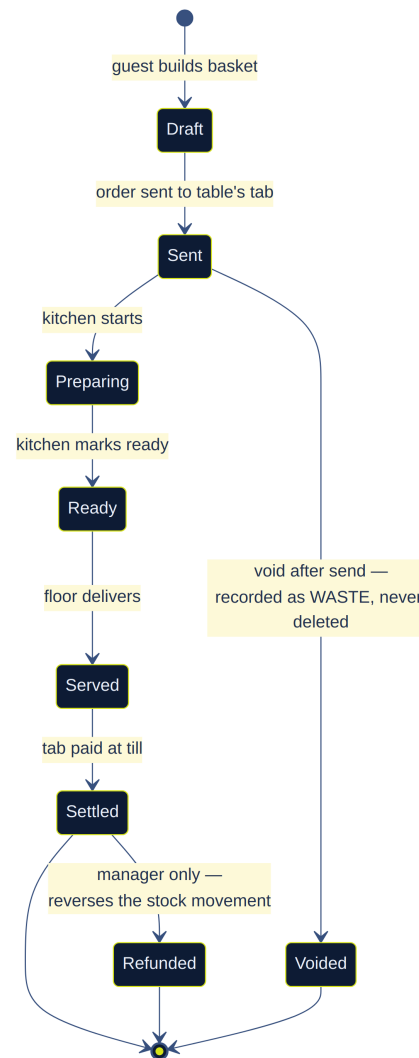


WHAT THIS SHOWS

No app, no account: the phone camera scans the table's QR code and the menu opens already bound to that table, in the guest's language. The order flows to the kitchen screen live, and status updates on the guest's page. Ordering is not paying — everything settles at the desk.

6 · Life of a Cafe Order

AUDIENCE: STAFF + TEAM

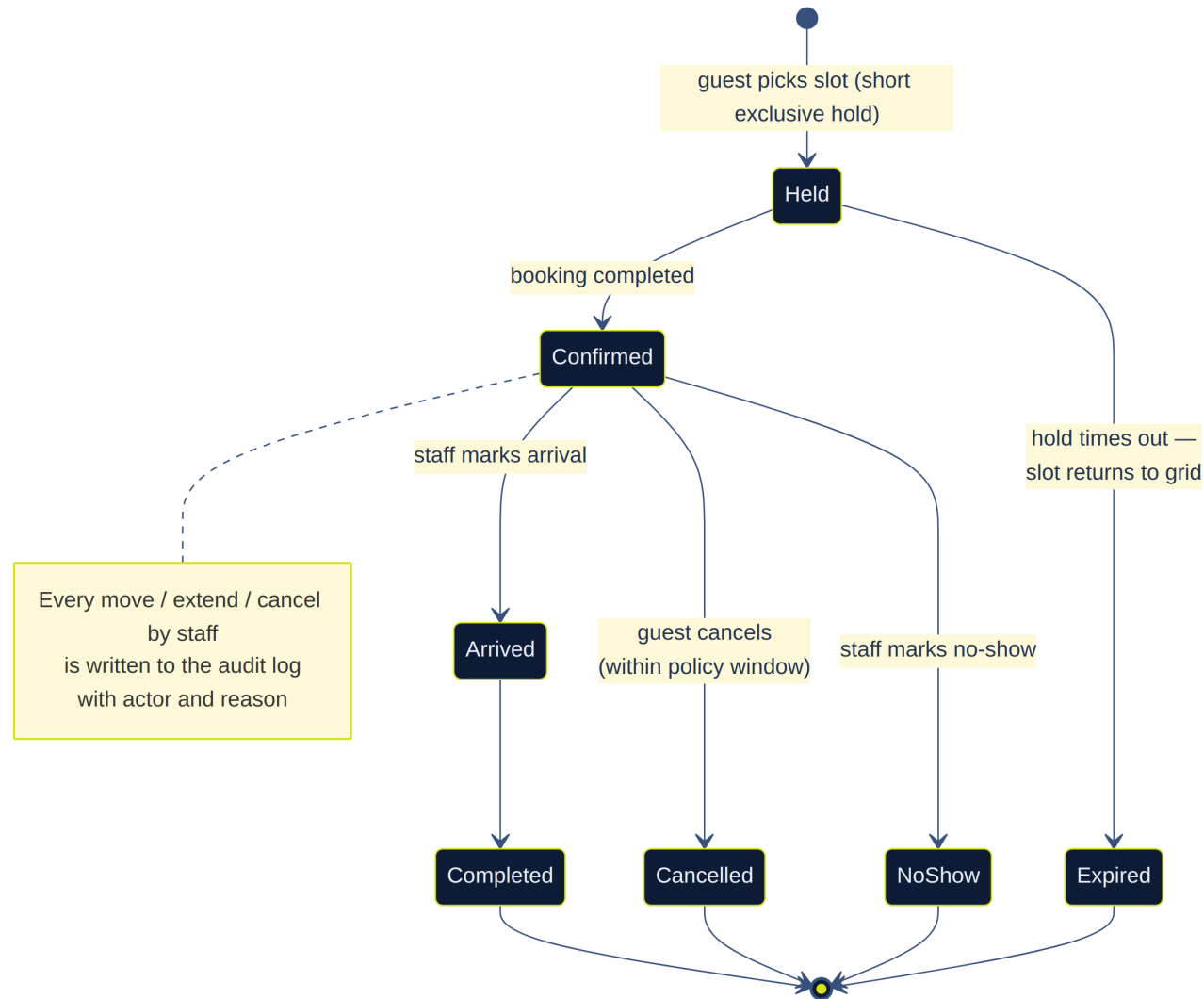


WHAT THIS SHOWS

Every order moves through defined stages. Nothing disappears: a void after the kitchen has the ticket is recorded as waste, and refunds (manager only) reverse the stock movement. This is what makes the day-close and variance numbers trustworthy.

7 · Life of a Court Booking

AUDIENCE: STAFF + TEAM

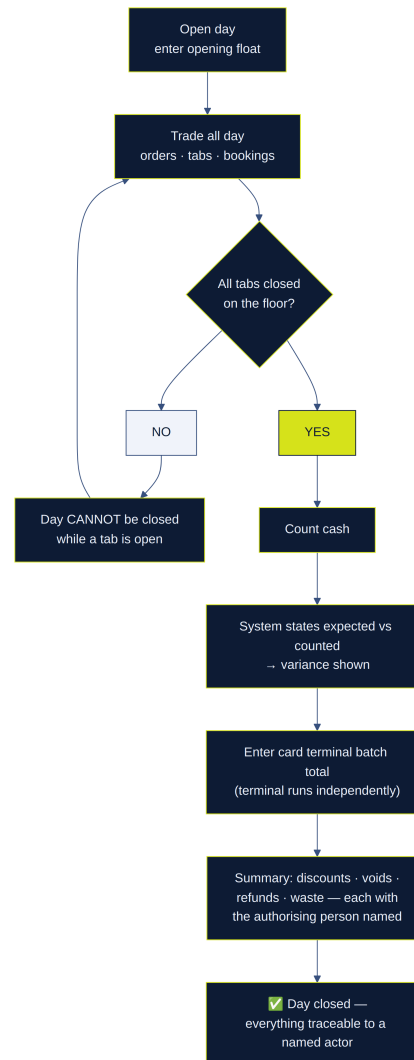


WHAT THIS SHOWS

From the short hold to completion, cancellation, or no-show. Every staff override — moving, extending, cancelling — is stamped into the audit log with who did it and why.

8 · Closing the Day

AUDIENCE: STAFF + MANAGER

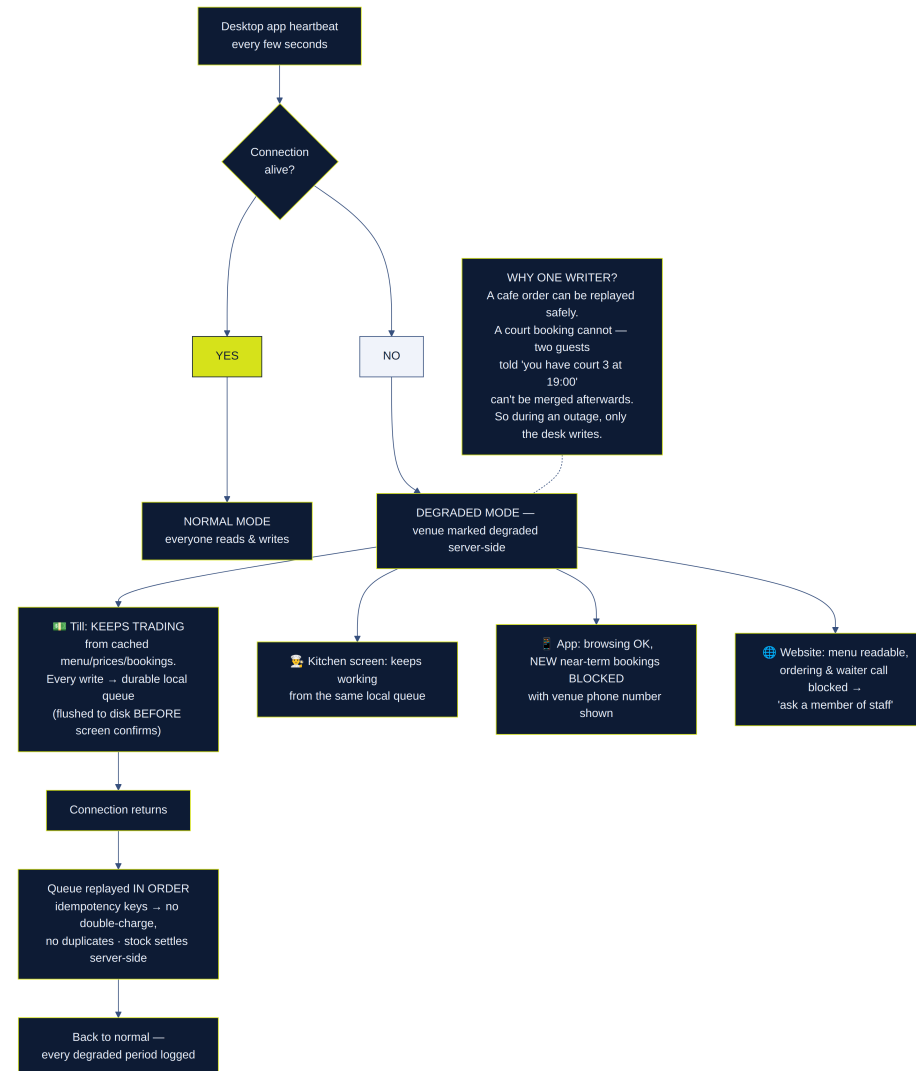


WHAT THIS SHOWS

The day cannot close while a tab is still open on the floor. Cash is counted against what the system expects, the card terminal batch is entered for reconciliation, and every discount, void, refund and waste entry is listed with the person who authorised it.

9 · When the Internet Drops — Degraded Mode

AUDIENCE: EVERYONE · KEY GUARANTEE

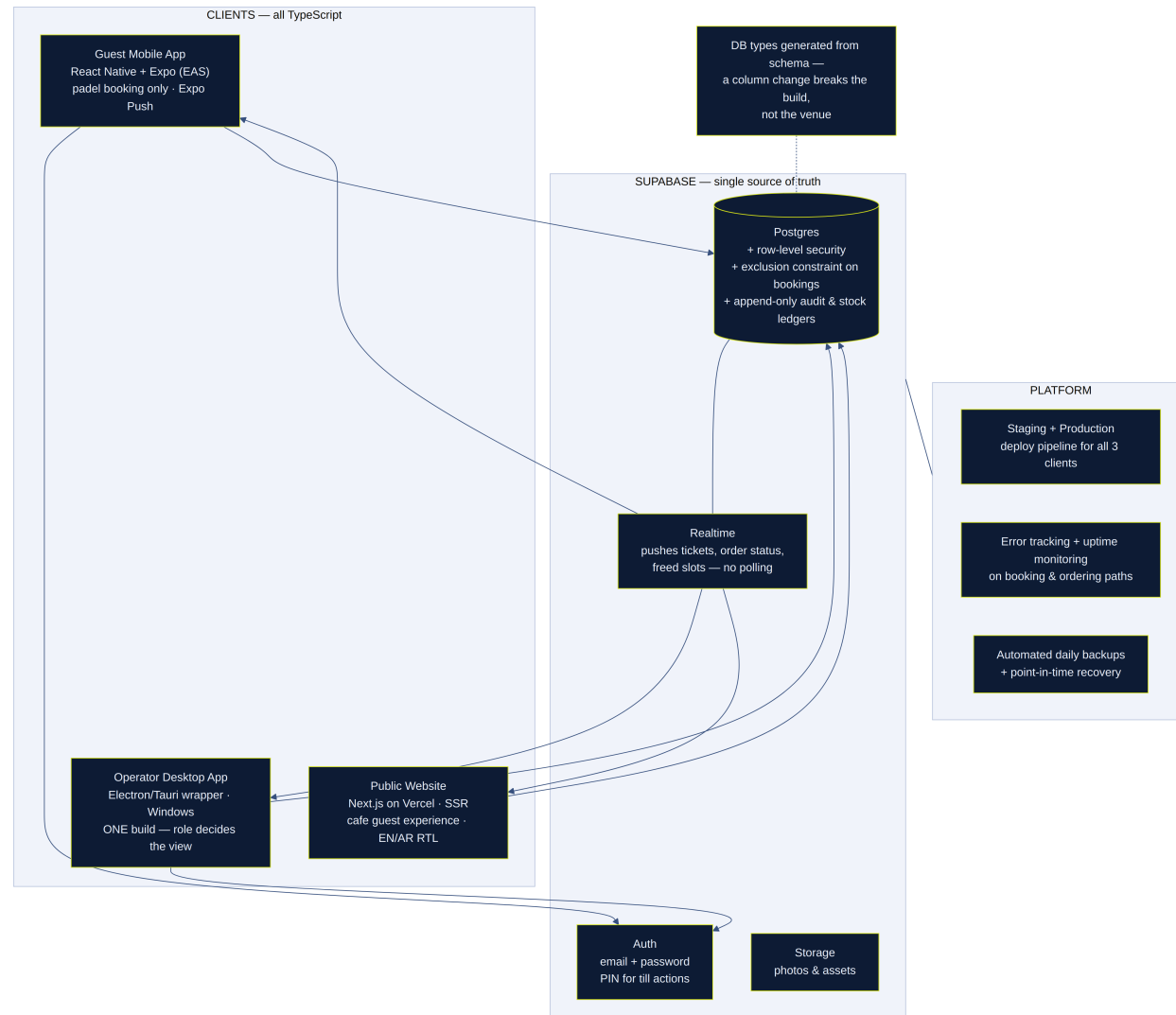


WHAT THIS SHOWS

The till and kitchen keep working from a local queue; nothing typed is lost even in a power cut. The app and website are blocked from near-term bookings so a slot can never be sold twice while offline. On reconnect, the queue replays exactly once and totals reconcile.

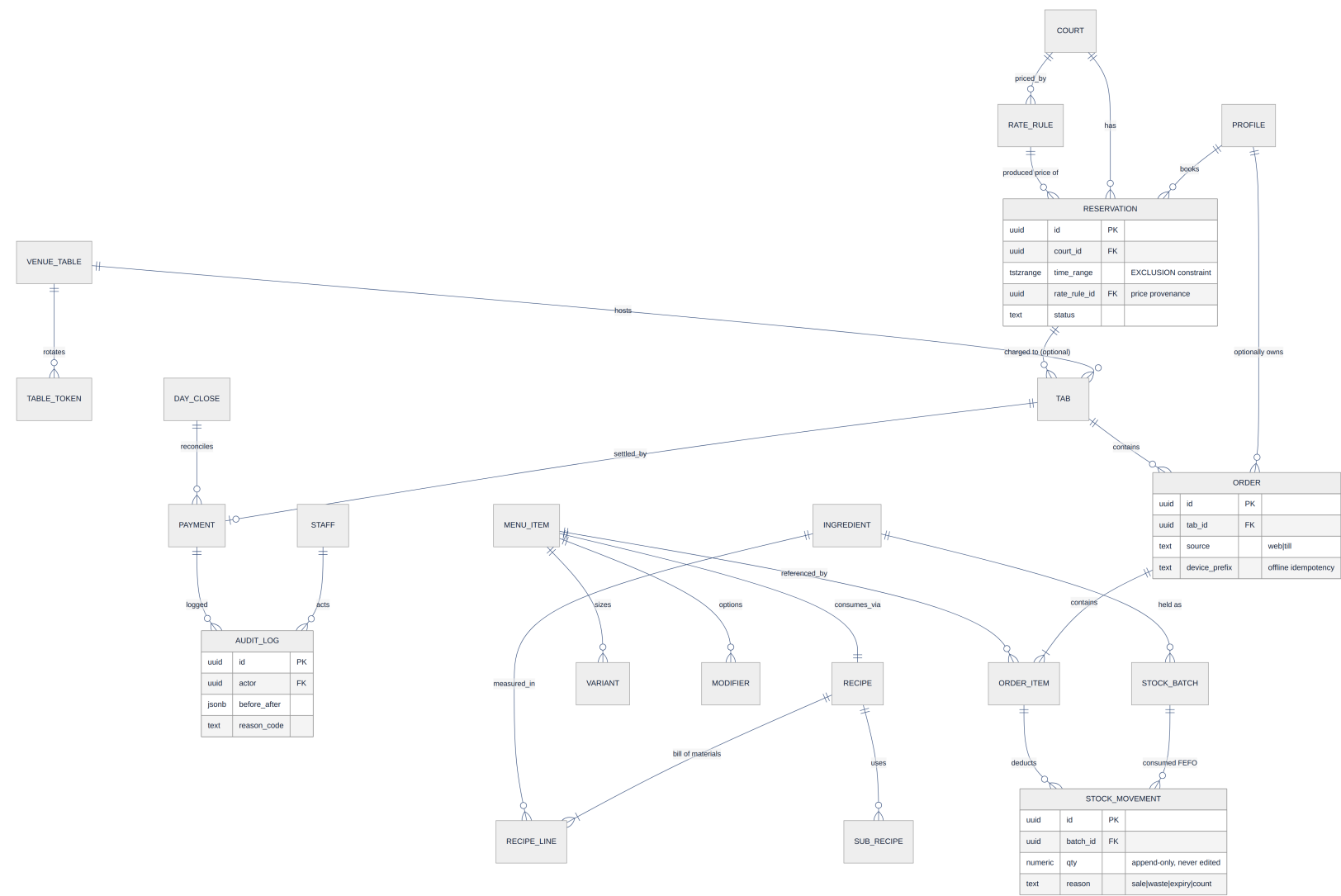
10 · Technical Architecture

AUDIENCE: TEAM



WHAT THIS SHOWS

TypeScript end to end. Supabase Postgres is the single source of truth with row-level security, the booking exclusion constraint, and append-only ledgers. Realtime channels push tickets, statuses and freed slots to every client. Types are generated from the schema, so a schema change breaks the build — not the venue.

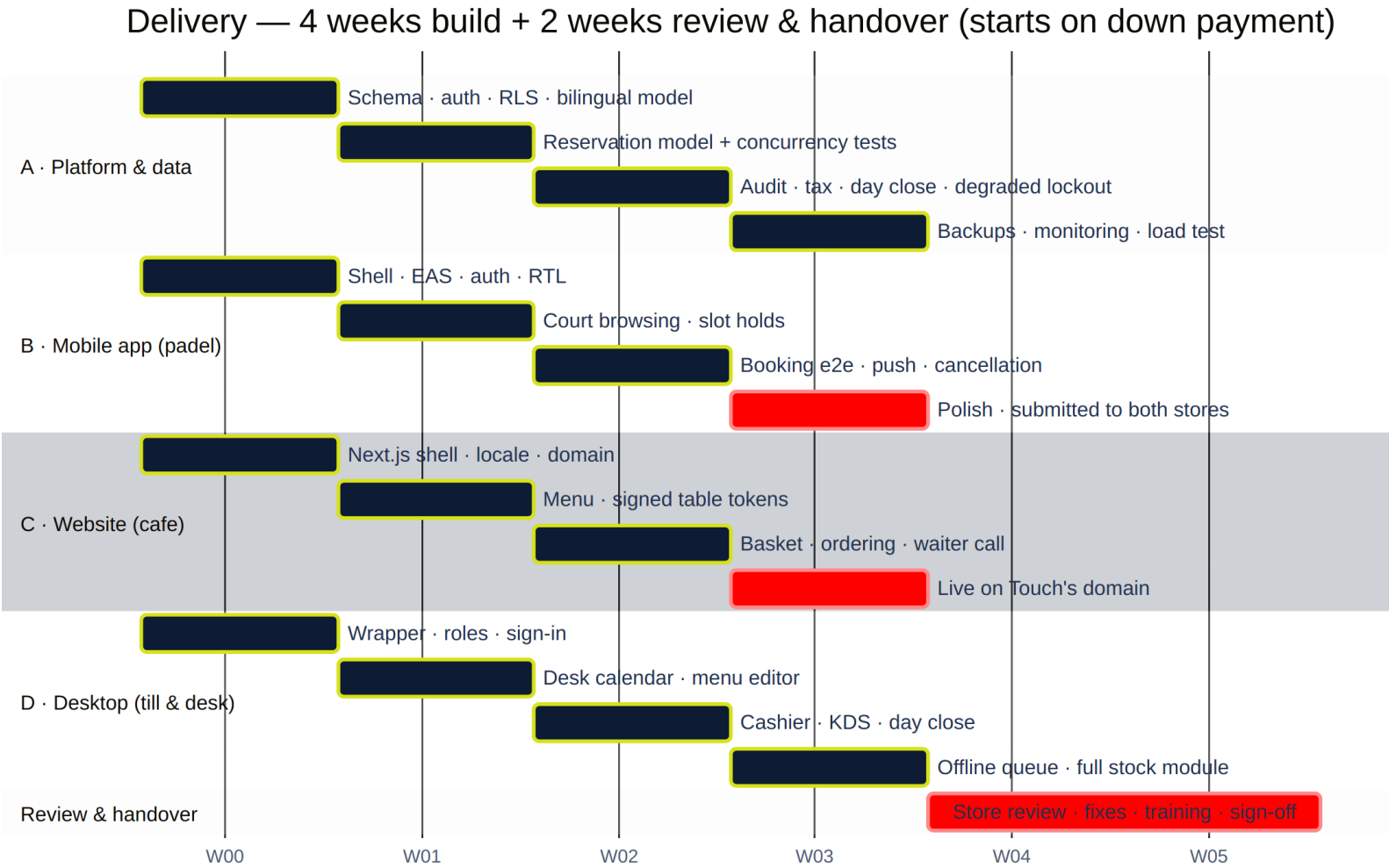


WHAT THIS SHOWS

The core entities and how they relate. Notable: one RESERVATION table with an exclusion constraint carries all court time; each booking stores the RATE_RULE that priced it; stock is an append-only ledger of movements against expiry-dated batches (first-expiring-first); a cafe TAB can be charged to a court booking.

12 · Delivery Plan — Six Weeks

AUDIENCE: EVERYONE



WHAT THIS SHOWS

Four parallel build tracks over four weeks, then two weeks reserved for store review, defect fixing, training and handover. Track A (platform) runs a few days ahead in week 1 — it is the foundation the other tracks build on. A working build is demonstrated at the end of every week.